STATEMENT OF WORK

Architectural-Engineering (A-E) Services Contract

Prepared by

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DAVIS-MONTHAN AFB
ARCHITECTURAL-ENGINEERING (A-E) SERVICES CONTRACT
STATEMENT OF WORK (SOW)

PART A INTRODUCTION

1.0 SCOPE

Provide Architectural-Engineering (A-E) Services as required by this Master Statement of Work for Air Force Bases, Air National Guard Bases, and related facilities in the greater Tucson and Phoenix areas, but primarily at Davis-Monthan AFB, Arizona. This statement of work (SOW) defines requirements for A-E services in support of Davis-Monthan's Facilities Sustainment, Restoration and Modernization (FSRM) program, and other areas of essential support.

The Contractor shall furnish the personnel, services, equipment, materials, off site facilities, and other requirements necessary for, and incidental to, the performance of work set forth under this basic contract, as well as each work assignment issued as a Task Order (TO).

Primary professional and technical services shall be performed by individuals who are credentialed members of architectural, planning, science, and engineering professions. Generally, a credentialed professional (a) is licensed (e.g., registered professional engineer) to practice in the state of Arizona and (b) commands the necessary expertise, in terms of knowledge and experience, to undertake the specified task to be identified at the TO level.

2.0 APPLICABLE DOCUMENTS

Comply with all applicable and current (1) federal, state, and local statutes, instructions, manuals, handbooks, regulations, guidance, policy letters, and rules including Unified Facilities Criteria (UFC), and (2) Presidential Executive Orders, in effect on the date of issuance of the TO. The Contractor shall be responsible for identifying and complying with all applicable requirements for each TO. The Contractor shall identify, use, and edit applicable Unified Facilities Guide Specification (UFGS). The following list includes some of the most common references for Department of War and US Air Force facilities/infrastructure projects:

- United Facilities Criteria (UFC), Specialty Specific, most current, https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc
- Unified Facilities Guide Specifications (UFGS), most current. https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs
- DAFMAN 32-1084, Standard Facility Requirements
- AFI 32-1020, Planning and Programming Built Infrastructure Projects
- Uniform Federal Accessibility Standards
- American National Standards Institute
- The Federal Acquisition Regulation (FAR), regarding Green Procurement; specifically, clauses FAR 23.202, FAR 23.403, FAR 23.703
- Davis-Monthan AFB Design Compatibility Guidelines (DCG)
- Davis-Monthan AFB Installation Facilities Standards (IFS) https://www.wbdg.org/ffc/af-afcec/installation-facilities-standards-ifs/davis-monthan-afb-ifs
- Davis-Monthan Contractor Environmental Guide

- 355th Communication Squadron - Telecommunications Installation Criteria for Facility Construciton and Renovaiton Designs
- UFC 4-010-06 and referencd Cybersecurity requirements

PART B ADMINISTRATIVE AND MANAGERIAL REQUIREMENTS

3.0 SECURITY ADMINISTRATION

The Contractor shall be responsible for understanding security obligations for the formulation of adequate regulatory procedures in the safeguarding of classified defense and other protected information, as required for a TO. Contract security requirements and Contractor access to classified information shall be in accordance with the "National Industrial Security Program Operating Manual (NISPOM) and as specified in the DD Form 254 - Contract Security Classification Specification specific to a TO.

3.1 Personnel Clearances

The CO may issue TOs whose scope of effort will be in support of classified programs. The Contractor shall be responsible for recruiting and retaining personnel with the appropriate clearances, as outlined in the individual TOs, to perform under this contract. Generally, all Contractor personnel assigned to work requiring access to any Government automated information technology system shall require a National Agency Check with Inquiries (NAC-I). Certain TOs will require the Contractor to work with documents with security classifications up to TOP SECRET. The Contractor shall identify Contractor employees on each TO to the appropriate Security Manager for verification in the Joint Personnel Adjudication System (JPAS) before TO performance is scheduled to begin. Specific access information shall be detailed in the DD Form 254 attached to the TO.

3.2 Document Accountability

The Contractor shall maintain accountability records for Contractor-generated and received classified material and comply with the provisions of the NISPOM. Contractor classified accountability record/receipts (AF Form 310, Document Receipt and Destruction Certificate) shall be used for accountability of classified material, to include transfer and destruction.

3.3 Task Order Security Requirements

The Contractor shall be responsible for complying with program office Security Classification Guides (SCGs) and shall be responsible for marking and protecting information in accordance with these same SCGs for each task containing or producing classified documentation. All Contractor employees shall observe and otherwise be subject to such security regulations as are in effect for the particular premises involved. When directed by the CO, the Contractor shall remove any employee who endangers national security, or is believed by the CO to be endangering national security. Removal shall be at no cost to the Government and an equivalent replacement shall be proposed and approved by the CO, as required, to complete task requirements.

All possible precautions and efforts shall be made to prevent the disclosure of any information that might provide prospective construction Contractors, manufacturers, or suppliers an advantage over others.

3.4 Industrial Security Office

All Contractor performance involving collateral classified information shall be under the security oversight of the Industrial Security Office (ISO), at the location. The appropriate security office providing oversight will be identified in individual TOs, as applicable. All Contractor performance involving collateral classified information off a military installation that is not inspected by the "local" customer's security agency shall be under the security oversight of the Defense Security Service (DSS).

4.0 SAFETY AND HEALTH

4.1 Worksite Safety & Coordination

Coordinate worksite activities to ensure the protection of human health and the environment the prevention of damage to property, utilities, materials, supplies, and equipment and the avoidance of work interruptions. Provide physical security to work area with security equipment and personnel as specified in each TO.

The Contractor must conduct work in compliance with DM's environmental policy, in accordance with the Environmental Management System (EMS), and respective to the significant aspects and environmental action plans (EAPs). The Contractor must comply with Occupational Safety and Health Administration (OSHA) safety and health regulations and local safety office requirements. The Contractor is required to provide the CO copies of any OSHA report(s) or host nation safety and health report(s) submitted during the duration of the TO.

4.2 Critical Issue Notification Letter

The Contractor shall notify the CO and COR of critical issues that may affect the contract performance and/or human health and the environment. The types of issues that require notification include, but are not limited to, health risks, spills, changes in critical personnel, and UXO. As an example, if UXOs were discovered during field activities, the Contractor would be required to immediately stop work, report the discovery to the base POC and COR, and implement the appropriate safety precautions. Commencement of field activities could not continue until clearance was received from the CO. On critical issues, verbal notification should be made immediately, followed by written notification as soon as practical.

4.3 Occupational Safety and Health Administration (OSHA)

The Contractor shall comply with OSHA safety and health regulations and local safety office (355 WG/SE) requirements. The Contractor is required to provide the CO copies of any OSHA report(s) submitted during the duration of the TO.

4.4 Worksite Coordination

The Contractor shall coordinate worksite activities with applicable existing base or area operations to ensure the protection of human health and the environment; the prevention of damage to property, utilities, materials, supplies, and equipment; and the avoidance of work interruptions.The Contractor shall coordination activities with the TO Project Manager and the following offices, as applicable:

Police/Security	Planning
Bioenvironmental	Utilities
Fire Department	Pass and Identification
Safety	Facility Managers
Environmental	Communications
Airfield Management	

4.4.1 AF Form 103, BCE Work Clearance Request:

The Contractor shall coordinate digging permits and bluestake requirements through the TO Project Manager and allow at least 21 calendar days for the project area(s) to be properly marked or cleared and to receive an approved AF Form 103. The Contractor is responsible for contacting Arizona Blue Stake (Arizona811) for non-Air Force owned utilities.

The Contractor must mark all areas that he wishes to have the Government bluestake with white paint PRIOR to the submission of the AF Form 103; and shall include a site drawing of the area to be bluestaked with the submission. Once a project site has been marked, it is the Contractor's responsibility to maintain the markings with stakes or whiskers.

4.4.2 Environmental

The Design and Contract Documents shall indicate the following:

The Contractor shall be required to meet or exceed the requirements and procedures of the most current Davis-Monthan AFB Contractor Environmental Guide in effect at the time of the TO contract award. A copy of this guide will be provided as part of the solicitation documents. This Guide includes procedures which if adhered to by the Contractor should ensure compliance with the Installation's environmental permits (as maintained by 355 CES/CEIE) as well as other regulations to include, but not limited to, training for the ISO 14001 Environmental Management System (EMS). To receive the initial EMS Awareness Training the Contractor will be required to register and login to The Environmental Awareness Course Hub (TEACH) at https://usaf.learningbuilder.com (must use Google Chrome). The detailed registration instructions are provided in Appendix E of the DMAFB Contractor Environmental Guide.

Additionally, the Contractor shall be held responsible for complying with all federal, state, and local environmental regulations included as part of the processes in designing the work included in the individual TO.
Contractor as required by individual TO shall obtain permits from the Pima County Department of Environmental Quality (PDEQ), or appropriate jurisdiction, Air Quality Division, for use of Contractor equipment which generate air pollution, for earth moving activities related to design investigations. Submit copies of permits to the Contracting Officer prior to beginning investigations. All permit costs shall be considered as having been included as a factor in the Contractor's TO proposal.

For projects involving asbestos, lead based paint, or mold testing submit copies of the test results to the Contracting Officer prior to the conceptual package submission. The Contractor shall be responsible for the proper management of testing waste per the DM Contractor Environmental Guide. All testing results shall be included within the design drawings and specifications. The Contracting Officer Representative will forward the plans and results to the installation's Environmental Office for review.

Contractors shall comply with all regulatory requirements/considerations if the TO requires soil disturbance during design investigations. These requirements may trigger Storm Water Pollution Prevention Plans (SWPPP), Air Activity Permits, and Natural and Cultural Resource protections. Area disturbance requiring permits are set forth by Arizona Department of Environmental Quality (ADEQ) and PDEQ. Treat areas subject to dust producing activities with liquid palliatives, which will not harm regrowth of vegetation, or other methods of dust control. Also, ensure methods to protect native plants, birds and wildlife are included in the design, where required.

5.0 PERSONNEL MANAGEMENT REQUIREMENTS

5.1 Contractor Personnel Performing Environmental Services

All Contractor personnel performing environmental services shall complete appropriate OSHA training and/or other applicable training related to the type of environmental services being performed as specified in individual TOs. Certificates shall be supplied at the TO level.

5.2 Ethics

The Contractor will be visible to the Davis-Monthan AFB community and other Government communities as a result of providing services. Consequently, the Contractor shall present a professional appearance and professional conduct.

5.3 Business Relations

The Contractor shall be proactive in identifying problem areas and providing corrective action. The Contractor and Contractor employees shall be customer-oriented and display a professional and cooperative attitude when performing service for the Government. The Contractor is responsible for providing timely notification to the Government of any acquisition or mergers involving the Prime Contractor to include the potential impact on this contract.

5.4 Professional Employees

Professional employees are defined by FAR Part 22.1102 as: "members of those professions having a recognized status based upon acquiring professional knowledge through prolonged study. Examples of these professions include accountancy, actuarial computation, architecture, dentistry, engineering, law, medicine, nursing, pharmacy, and the sciences (such as biology, chemistry, and physics and teaching). To be a professional employee, a person must not only be a professional but must be involved essentially in discharging professional duties."

Primary technical services shall be performed by individuals who are credentialed members of architectural, engineering, planning, and science professions. Disciplines include, but are not limited to, architects, landscape architects, surveyors, interior designers, engineering disciplines associated with facility design and construction, environmental engineers and scientists, cost estimators, specification writers, Computer Assisted Design and Drafting (CADD) technicians, publication writers/editors, technical writers, graphic artists, Geographic Information System (GIS) analysts, planners, urban designers, economists, financial analysts, and housing market analysts. A credentialed professional is licensed (e.g., Registered Architect, Professional Engineer) to practice in the state where a facility is located, and has the necessary expertise, in terms of knowledge and experience, to accomplish specific tasks identified in the SOW.

5.5 Key Personnel

Personnel identified in your SF 330 A-E Qualifications package are considered key personnel and shall be the personnel utilized to accomplish TO requirements. Substitution of key personnel is only authorized if the proposed individual becomes unable to work or leaves the company. If one of these situations occurs, the Contractor shall replace the employee with an employee with the same or higher credentials, education, and experience. All substitutions of key personnel shall be approved by the CO prior to placing the individual on the TO.

5.6 Management Structure

The Contractor shall implement and maintain a management structure consistent with proven methods of management of diverse technical workloads under a TO system. The structure shall be capable of responding to requirements across the full range of workload potential and be adaptable enough to meet long term/stable tasks and short notice, short duration tasks. The structure must be flexible enough to allow rapid reassignment of personnel, as well as, the assignment of personnel to multiple, sometimes diverse tasks. Lines of authority and responsibility shall be clearly established and delineated for managerial, technical, administrative and support personnel. Administrative and support functions shall also be structured to be responsive to TO requirements.

5.7 Program Management

The Contractor shall employ a program management structure to ensure the efficient execution of all TOs and the capability to report on the status of work performed. The Contractor shall use a single Program Manager (PM) with the authority to administer all project activities and serve as the principle Point of Contact (POC) for all matters regarding project administration and reporting. To ensure efficient management and administration of all TOs, the Contractor shall also provide an Alternate PM. The PM or Alternate PM shall provide the direct, recurring interface between the Contractor, the Quality Assurance personnel, and the applicable directorate staff. The Contractor shall promptly notify the CO of any issues or problems requiring a Government response.

5.8 Project Management

The Contractor shall identify a list of PMs and Alternate PMs for this contract and identify a PM and an Alternate PM for each TO. This individual shall sign status reports for the TO and oversee the overall task through completion.

6.0 CONTRACT DOCUMENTS

6.1 Key Personnel Roster (KPR)

The KPR shall be submitted in electronic format, with response to the Prime Contract RFP. Subsequent submission to maintain a current KPR shall be submitted electronically via e-mail to the CO within **10 workdays** of a personnel change e.g., employee quit, employee transferred, etc. All substitutions of key personnel shall be pre-approved by the CO prior to adding the employee's to the KPR. Proposed substitutes shall have equivalent certification(s), qualifications, and experience as the individual he or she is replacing. Requests for personnel additions and/or substitutions shall be submitted, in writing, to the CO **14 workdays** prior to requested start date for the employee, if practicable.

6.2 Labor Rate Schedule

The negotiated Fully Burdened Labor Rate (FBLR) for each labor category, shall be incorporated into the Prime Contract as the "*Labor Rate Schedule*" **Attachment 1**.

6.2.1 Calculating FBLRs

FBLRs for each team member shall be established at award. The FBLR is based on the employee's current "base rate" with currently approved indirect rates and profit applied to calculate the FBLR.

The methodology used to calculate each employee's FBLR shall be as follows, or shall be a formula / calculation approved by DCAA that is in accordance with acceptable accounting procedures and is consistently used by the firm:

Base Rate	Fringe	OH	Subtotal #1	G&A	Subtotal #2	Profit	FBLR
	0.00%	0.00%		0.00%		0%	
$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00

6.2.2 Indirect Rates

Contractors shall provide sufficient cost data to the CO for evaluation of cost pools, accumulating practices, and allocating G&A costs proportionally among contracts based on dollar value for indirect expenses. Sufficient documentation shall be provided to support indirect rates at award for the projected 60 months of the performance.

Indirect rates (Fringe, Overhead, G&A, C.O.M., etc.) proposed at award shall be determined reasonable to be awardable. High indirect rates will be viewed as inefficient cost control by the firm and considered unrealistic / unreasonable and not awardable unless negotiated to a reasonable rate. It is highly recommended that a serious review of corporate expenses, cost pool allocations / accumulation, unallowable costs, etc. be accomplished to ensure the indirect rates proposed to the Government are adequately supported and determined reasonable for award. Supporting documentation for the indirect rate percentage proposed shall be provided. Supporting documentation shall include Defense Contract Audit Agency (DCAA) approved rate documentation, if preformed and cost accumulated in each indirect cost pool how the percentage applied to the base rate is formulated and how the appropriate indirect cost(s) are proportionally distributed among all contracts as required by cost accounting practices. With close review of indirect rates mandated, it is in the best interest of the firm to conduct a thorough review of their expenses and accumulating processes prior to submitting data for review by the Government.

6.2.3 Audit of Records

A specific field audit shall be requested of DCAA prior to initial contract award in accordance with FAR 52.215-2. However, in the event the audit cannot be performed in a timely manner due to DCAA's workload, to ensure continuation of required mission essential services, the CO may negotiate fair and reasonable rates based on available information provided by the Contractor pending completion of the requested DCAA audit. At the completion of the DCAA audit of indirect rates after negotiated rates are in effect, if the audit results find the Contractor overstated the indirect rates used to calculate the FBLRs the Contractor shall reimburse to the Government the percentage of overpayment for all work performed from the effect date of the initial contract award or any redetermination thereafter to the date the rates are modified to incorporate the reduced indirect rate percentage.

6.3 Quality Control Plan (QCP)

The Contractor shall submit a QCP, with submission of the Prime Contract proposal for review and approval by the CO. The plan shall identify the actions management shall employ to monitor performance and ensure the quality of deliverables in accomplishing contract and TO requirements and governing directives, regulations, and policies. The plan shall be updated as needed to ensure quality service and deliverables are maintained throughout the contract performance period. CO review and approval is required for all revisions.

7.0 QUARTERLY REPORTING

7.1 Quarterly Status Report (QSR) - Prime

Prepare and submit via e-mail to the COR and CO a QSR in the Contractor's format. The QSR shall included, at a minimum, a brief description and status of current TOs, TOs completed during the quarter, and current issues with proposed solutions.

8.0 ANNUAL REPORTING / UPDATES

8.1 Standard Form 330 A-E Qualifications (SF 330)

The Contractor, at a minimum, shall submit and update their SF 330 qualifications package annually via e-mail to the CO. The Contractor may submit an updated SF 330 at any time prior to the annual requirement when changes occur in the firm's qualifications, personnel, or experience (i.e. TOs performed, subcontractors added, etc.) that the firm feels would better represent their firm's current qualifications/capabilities to the board when the board reviews the firm's SF 330 for ranking of firms for issuing TO Request for Proposals (TO RFPs).

PART C TASK ORDER AWARD PROCEDURES AND REQUIREMENTS

9.0 TO MANAGEMENT REQUIREMENTS

9.1 Design Schedule Requirements

9.1.1 Project Planning Chart (PPC)

Provide schedules for tracking TO work progress. Following the TO contract award, the Contractor shall finalize and submit their proposed Design completion schedule. The Contractor shall prepare the schedules using the government provided PPC worksheet. The Contractor shall submit for government review and approval the PPC no later than ten (10) calendar days following issuance of the Notice to Proceed. All tasks required under the TO shall be included in the PPC.

9.2 Material Approval Submittal (AF Form 3000)

The AF Form 3000, *Material Approval Submittal* shall be used for submitting both Government approved and information only submittals in accordance with the instructions on the reverse side of the form. The Contracting Officer shall furnish these forms to the Contractor. This form shall be properly completed by filling out the heading blanks and identifying each item submitted. Special care shall be exercised to ensure proper listing of the specification paragraph and/or sheet number of the contract drawings pertinent to the information submitted for each item(s).

The Contractor shall forward all material submittals required by the technical specifications using the AF Form 3000 as stated in paragraph b1 above. The Contractor provides the submittals in the number of copies as specified on the AF Form 66, *Schedule of Material Submittals* and one original of the completed AF Form 3000 to the Contracting Officer no later than the required submission date specified on the AF Form 66 or 10 calendar days from the Notice to Proceed, whichever is later.

Identify each submittal by project, Contractor, drawing or detail number, and specification section number, as appropriate. Mark submittals to show the specific item(s) that will be furnished. Include intended use of the submittal and any other pertinent information necessary for a complete evaluation. If the Contractor provides a submittal which show variations from the contract documents, the Contractor shall completely describe such variations in writing clearly and separately from any other portion of the submittal. Highlighting or marking in any manner the proposed variations is not sufficient to comply with this requirement. This requirement applies irrespective of the provisions of any specification which follow this section and which may be particular to specific item(s).

9.3 Design Schedule Of Material Submittals (AF Form 66)

A draft AF Form 66, *Schedule of Material Submittals*, for the Design only project, will be provided by the Government in the TO. The Contractor shall submit an updated AF Form 66, for the TO; no later than ten (10) calendar days following issuance of the Notice to Proceed.

9.4 Geospatial Mapping Standards

All products associated with this contract that provide a map representation of the location of installation features (historical, existing, or planned) including installation maps, site plans, area development plans, walls-out as-built depictions, or other related overhead (plan) views of an installation (partial or entire) must adhere to the following requirements. (NOTE: This requirement does not currently involve walls-in facility floor plans or interior renderings.)

All maps and associated data must comply with the latest version of Spatial Data Standards for Facilities, Infrastructure, and Environment (SDSFIE) available from the SDSFIE web site. The SDSFIE web site address will be provided under separate cover. These data will be organized using the latest version of the SDSFIE standard, Air Force Adaptation, approved by the Installation Geospatial Information & Services (IGI&S) Governance Group (IGG) available from the Air Force Civil Engineer Center (AFCEC) Geo Integration Office (GIO) as the USAF lead for IGI&S capabilities. The SDSFIE standard is the authority for file and feature class identification and definition, attribution and valid domain values. When any geospatial information collected as a result of the contract includes information identified in the Common Installation Picture (CIP) or recognized Mission Data Set (MDS) the Contractor will deliver data consistent with the established requirements for the data and will ensure functionality with the receiving system. Information must be collected at no less than 1:1200 scale for base cantonment areas and 1:4800 scale for larger undeveloped base areas. Spatial data will meet or exceed National Map Accuracy Standards at those scales. Metadata will be provided and will use Federal Geographic Data Committee (FGDC) Content Standards for Digital Geospatial Metadata (CSDGM) for organization.

Geospatial data must be delivered in a geo-referenced Geographic Information (GIS) format (feature-based file structures with one-to-one cardinality between spatial records and attribute records) using Environmental Systems Research Institute's (ESRI) geodatabase format. All attribute data as specifically outlined in the task order contract must be included either in the GIS data file or as a separate table with a SDSFIE key variable that may be used to relationally join the separate table with the GIS data file. All geospatial data must be delivered using the projection, coordinate system, and coordinate units stipulated by the AFCEC GIO (e.g, WGS-84, North American Datum 1983 (NAD83) projection, State Plane Coordinate System, feet or metric coordinate units). Further guidance on mapping units, coordinate systems and projections is available from the AFCEC GIO and/or Installation GIO.

Mapping- or Survey-Grade Global Positioning Systems (GPS) or comparable traditional survey methods will be used to collect geospatial data. The use of mapping- or survey-grade GPS will depend on the precision requirements of the product data. These requirements will be specified later in this SOW for all contract activities where geospatial data are involved. In the case of contracts involving utility construction, location and attribute data will be obtained at the time of excavation. Further information about precision requirements should be obtained from the AFCEC GIO or installation GIO.

Source data and product data remain the property of the US Government. The Contractor may be required to explain and demonstrate the company's process for protecting all geospatial data, including but not limited to geometry, attributes, metadata, topologies, and relational database schemas and operations used in association with TOs. The Contractor may be required to sign a non-disclosure agreement attesting to the same before source data are released. Further information about security and nondisclosure requirements should be obtained from the AFCEC GIO and/or installation GIO. Some installation map data, source and/or product, may be considered by the government to be “sensitive, but unclassified” or “Critical Unclassified Information (CUI)”. The intent of this clause is to prevent intentional or unintentional dissemination of “sensitive, but unclassified” information to include unauthorized access to the source and product data by any entity wishing to do harm to the USAF or U.S. Government while the data resides on the Contractor's computer network. The Contractor is not authorized to release this information to any third party without the explicit consent of the Air Force Civil Engineer Center (AFCEC) or the involved installation. All source information must be returned to the government POC or destroyed upon completion of the project. Special requirements for handling classified map data, if applicable, will be addressed within individual TOs. Final deliverable geospatial data must be provided to the installation GIO in an electronic, digital format according to specifications provided above.

10.0 INVOICING

10.1 Cost and Status Reporting

10.2 Contractor’s Progress, Status, and Management Report (CPSMR)

Prepare and submit TO CPSMRs. The CPSMR shall be used to review and evaluate the overall progress of the TO project, along with any existing or potential problem areas. The CPSMR shall include a summary of the progress, events that occurred during the reporting period, discussion of performance, identification of problems, proposed solutions, corrective actions taken, and outstanding issues. Cost information shall be included in this report.

The Contractor shall prepare scheduled progress reports on the PPC. The Contractor shall submit for government review and approval the government provided worksheet at each scheduled progress report date (provided by the Contracting Officer). These reports are used to track the schedule and progress of the project throughout the life of the contract; and are directly related to invoice evaluations.

11.0 GENERAL INFORMATION

11.1 Place of Performance

Will be identified in individual TOs.

11.2 Government Points of Contact

Will be identified in individual TOs.

PART D PLANNING SERVICES

The Contractor shall provide the following services as specified in individual TOs:

12.0 INFRASTRUCTURE AND INSTALLATION PLANNING AND PROGRAMMING

Develop, update, integrate, publish and present planning and programming services as specified in the individual TOs. Develop, update, integrate, publish and present General Plans, Master Plans, Comprehensive Plans, Asset Management Plans**,** Area and Installation Development Plan(s) (ADP/IDP), facility use survey (FUS), space utilization plan(s) (SPACE-UP), and other plans as specified.

12.1 Planning and Programming

Services include support to establish or sustain infrastructure programs; develop plans and programs for future funding and execution of requirements.

12.1.1 Planning Actions

Develop a short-term and long-range plan of action to achieve compliance with the proposed mission, including regulatory requirements to satisfy initiatives necessary to acquire the authority and/or resources to accomplish the mission's planned work as specified in individual TOs. Review available documentation and develop criteria to prioritize requirements, analyze projected projects, provide execution options (funding release dates, obligation schedules and Notice to Proceed milestones), and accomplish other similar recommendations.

12.1.2 Program Management Integration

Develop, present, and publish the installation and headquarters/command level planning, programming and budgeting submissions in support of the military's force structure, associated installation programs, and related projects as specified in individual TOs. Assist with the development of a master schedule to execute mission support programs.

12.1.3 Programming Actions

Provide documentation necessary to acquire the authority and resources to accomplish MILCON, MFH, and SRM work. Prepare URDs utilizing project management plans; data gap identification; project definition; drawings; space analysis; DD Form 1391 development; and parametric cost estimate as specified in individual TOs. In the event of discrepancies between conditions described in the contract and actual on-site conditions, document the discrepancies on a Site Survey Report.

12.2 Base Capacity Analysis, Studies, and Reports

Prepare analysis of base carrying capacity to support facility use requirements, including airfield/parking-ramp capacity studies; facility utilization assessments and study report(s); utility infrastructure; explosive safety quantity distance criteria; base environmental condition report(s); environmentally sensitive site analysis; report(s) on the presence of endangered species and their habitats; and report on natural and cultural resource survey(s) and supporting impact mitigation(s) as specified in individual TOs. In the event of discrepancies between conditions described in the contract and actual on-site conditions, document the discrepancies on a Site Survey Report.

12.3 Comprehensive Planning Program

Review documentation to establish a systematic framework of decision making for development of an investment strategy for the physical, real property assets of the Government and related environmental programs as specified in individual TOs. Prepare BCPs as specified. Data gathering in support of current or future BCP efforts shall include review of existing data, previous studies, plans and ortho-rectified and

LiDAR aerial photography and as necessary replace any outdated data and aerial photography with updated information and aerial photography in the format specified in the TO.

12.4 Infrastructure Assessment Study

A multidisciplinary team of engineers and planners shall assess base or facility level infrastructure capabilities to determine conditions to meet current or future needs. Prepare an Infrastructure Plan document that describes, assesses, and analyzes the infrastructure systems as specified in individual TOs. The plan includes utility systems, communications systems, navigational aids, and fire protection measures as they apply to installation development. The study shall provide a narrative and graphic description of each infrastructure system and subsystem, to include location data, capacities, peak demands, and condition. The plan shall provide recommendations to correct existing or future deficiencies/shortfalls. The plan shall identify, prioritize, and cost infrastructure projects. Data gathering for present or future Infrastructure Assessment Studies shall include mapping of each utility system at each base, the locating and identifying of visible and underground utility infrastructure and delivery of them in a GIS format. The following systems and subsystems shall be addressed and analyzed, as applicable. Others may be added as necessary.

Utility systems	Communications systems
Water supply and distribution system	Information transfer system
Sanitary sewerage system	Telephone switching system
Storm drainage system	Data communications
Electrical distribution system	Long haul communications
Central heating/cooling system	Radio systems
Natural gas system	Navigational aid (NAVAID) systems
Liquid fuel system	Fire suppression systems
Cathodic protection system	Fire Protection systems
Industrial waste system	Alarm system

12.5 User Requirements Document (URD)

Prepare URDs utilizing expert investigation and analysis that produce project management plans; data gap identification; project definition; drawings; space analysis; alternatives development; formal comparative (economic) analysis; DD Form 1391 development; and parametric cost estimates. A URD is documentation of a programming charrette for future project(s). A programming charrette is a planning technique where a team of contracted professionals coordinate with using agencies and other functionals to identify space requirements, detailed costs, and other project support documentation. These charrettes are critical to validate and approve projects before they are submitted to USAF, or other approval or reviewing entities. Specific deliverables will be identified in the specific TO.

12.6 Land Use Planning and Analysis

Analyze land use data to determine the effect of proposals on existing and proposed land use plan(s); analyze land use documents and plan(s) to determine current and future land use on/near installations and facilities; analyze plan(s) and studies of future land use proposals; determine the effect of proposals on installations, local, and regional land use requirements; identify and recommend planning measures necessary to overcome problems identified; revise/update land use plan(s) in support of the BCP for installations and ranges as specified in individual TOs. All ROD documentation for current and anticipated land use shall be consistent with the AF policy and guidance.

12.7 Sustainable Planning and Program Management

Utilize a multidisciplinary team of planners, environmental specialists, designers and construction specialists to develop a Sustainable Area Development Plan (SADP) for existing and new missions on all installations and other locations of interest as specified in individual TOs.

12.8 Transportation Planning and Analysis

Support the analysis and development of transportation systems that improve material supply efficiencies and that reduce the cost of base transportation requirements as specified in individual TOs. Obtain data or utilize existing data to determine the effects of proposed projects, mission changes and/or reuse of installations and/or facilities on local and regional traffic systems.

12.9 Landscape Development

Provide landscape development planning, design, and sustainability concept plans, and design documents contract packages as specified in individual TOs. The submission shall include:

a. Site Analysis
b. Site Design
c. Design of Circulation Systems
d. Common Area Design
e. Planting Design
f. Forestry
g. Site Furnishings
h. Anti-terrorism/force protection
i. Grounds Maintenance Cost-reduction Opportunities

12.9.1 Development Plans

Provide landscape development plans to implement water-efficient landscaping (xeriscape), making use of native plants and grasses, inert groundcovers, and efficient sprinkler systems.

Plans shall include:

a. Irrigation scheduling for landscape areas,
b. Inspection of existing irrigation system effectiveness,
c. Proposed retrofit for inefficient irrigation systems (e.g., in-ground versus hose-end sprinklers) and cost estimate,
d. Proposed landscape development plans and cost estimate, and
e. Guidelines for water-efficient design criteria for future systems.

12.9.2 Sustainability Concept Studies and Plans

Provide all concept studies, design plans and contract documents required to develop and implement projects for improving sustainability and reducing maintenance cost. The study shall include all opportunities within the study area and factors influenced by the surrounding area.

Subjects for analysis include:

a. Site Design
b. Circulation Systems

c. Common Area Design
d. Planting Design
e. Forestry
f. Irrigation Design
g. Site Furnishings
h. Site Security
i. Landscape Maintenance and Management
j. Cost estimate as identified in individual TOs

12.9.3 Force Protection Concept Studies and Design Plans

Provide all concept studies, design plans and contract documents required to implement projects for improving anti-terrorism/force protection within the exterior site/study area.

The study shall include:

a. Site selection
b. Area development planning
c. Vehicular site design
d. Standoff zones
e. Orientation of buildings on the site
f. Relationship of roads
g. Landforms and natural resources
h. Physical barriers
i. Landscape planting
j. Parking
k. Site furnishings
l. Barriers
m. Fencing
n. Walls
o. Gates
p. Planters
q. Natural features
r. Berms
s. Ditches and
t. Cost estimates

12.10 Advance Planning

Perform advanced planning services as specified in individual TOs such as:

a. Requirements and Management Plans
b. Fact finding studies
c. Visual analysis
d. Facility condition/utilization assessment
e. Site, asset management services, environmental, traffic and/or facility studies and analysis
f. Project criteria development and analysis
g. Comprehensive plan preparation
h. Sub-area development plans
i. Multi-year facility maintenance and repair plans, and other pre-design investigations
j. Complete related field investigations and research as required.

12.11 Community Planning and Sub-Area Development Plans

Perform required field, planning, and facilities requirement studies as specified in individual TOs. Provide long-term improvement/replacement plans with associated development scenarios with alternatives (preferred and non-selected) and economic/feasibility planning documents. The documents will be used as a tool to aid installations and facilities with their program planning and execution.

12.12 Dormitory Master Planning

Develop a Dormitory Master Plan to be used for future project programming and execution. The objective of the DMP is to perform detailed analysis of unaccompanied enlisted personnel housing (UEPH) and to provide the Air Force with a comprehensive investment tool for future year programming. The Contractor shall perform complete building condition and functional assessments and develop recommendations, costs, and scoring. The recommendations for existing dormitories should include renovation versus replacement based on the condition and functional assessments, force protection requirements, and cost. Site Analysis and Dormitory Campus Area Development Plans for each installation should be developed as part of the overall DMP. The plans should include campus infrastructure requirements, renovation vs. replacement, future facility siting, and phasing recommendations.

12.13 Energy Studies and Plans

Develop comprehensive energy analysis for buildings, areas of the base or the whole base depending on the requirements of the TO. The contractor shall perform energy audits, develop recommendations and provide detailed calculations and reports on energy analysis findings.

PART E TRADITIONAL SERVICES (SRM)

13.0 TITLE I SERVICES

The A&E shall perform all services related to a specific construction project required to investigate, design and to furnish complete construction documents consisting of contract drawings, technical specifications, design analysis, cost estimate breakdown, material submittals, and addenda as required, for the accomplishment of each project in accordance with requirements of the statement of work. Title I services may include all aspects of design such as preparation and/or review of contract plans, specifications, scheduling, cost estimates, building commissioning services and preparation of operating and design manuals. Title I efforts also encompass those efforts required to support and develop design work, including planning and programming, program management, project scoping, studies, investigations, evaluations, consultations, conceptual design, Building Information Modeling (BIM) employment, design analyses, comprehensive interior designs, interior facility functional/relational offices compatibility studies, value engineering, topographic survey services, and infrastructure systems.

Specific design requirements will be identified at the TO level. The A-E Contractor will utilize the Design UFC 1-300-09N for design requirements. A design analysis will be accomplished for each phase/percentage of the design process to address all pertinent facility systems. Additional and more detailed guidance for each discipline is found in other Unified Facilities Criteria.

Submittals will be considered incomplete and immediately disapproved if all elements listed in each submittal phase below are not included with the submittal, unless such items have been waived by the Contracting Officer in writing prior to the submittal being made to the Government. Following each design submittal the Government will be given time to review the submittal and provide any comments to the Designer. Following the Government review of each submittal the Designer shall be required to attend a Design Review meeting. The Designer shall be responsible for conducting/running all Design meetings.

13.1 Scope of Work

Perform all Title I design services, as described herein and incidental work as necessary:

Provide complete (100%) Design Services as described in TO, including preparation of complete Bidding and Construction Documents meeting Government standards for competitive bidding.

The Government reserves the right to request major changes during / through the Preliminary design phase that are within the original scope of work and lesser changes thereafter at no additional cost. Major changes requested by the Government after the Intermediate design phase or changes to the scope of work will be subject to negotiations for equitable compensation to the Contractor. No "change work" shall be performed until these negotiations are concluded and the TO is modified.

Other changes necessary to bring the project cost within budget or to bring the design into compliance with governing codes or regulations or DEQ requirements are the responsibility of the Contractor and are not justification for additional compensation at any design stage.

The design submittal provides a complete and final working set of contract documents ready for bid solicitation and ready for construction.

13.2 Investigative Services

In support of the studies, reports, and design of construction projects, the Contractor is directed to accomplish some or all of the following investigative services as detailed in the individual delivery orders.

Perform a survey of the project site that may include topography, cross sections, drainage survey, bench level circuit, and soil boring data. A registered professional land surveyor shall required for all surveys used by the Government. The Contractor shall use conventional surveying and other methods, such as a Total Stations, Global Positioning System (GPS) for field data collection. A separate electronic data deliverable is required for utility information and shall be a CD-ROM in a format that conforms to the latest version of both the CADD/GIS Technology Center’s Spatial Data Standards and A&E/C CAD standards. These standards can be found at https://cadbimcenter.erdc.dren.mil/. Data delivered in a format other than ESRI geodatabase must have an external spatial reference (.prj) file attached that specifies the parameters of the coordinate system in standard ESRI format.

13.2.1 Asbestos Survey And Report

Verify potential ACM sources using Bulk Asbestos Sampling Techniques in accordance with AHERA (Asbestos Hazard Emergency Response Act) guidelines and Polarized Light Microscopy (PLM) analysis. Take representative samples throughout the building (including basement, first floor, second floor, spaces above the suspended ceiling, accessible crawl space and attic) that may be demolished or that may be penetrated or otherwise disturbed during renovation of all potential ACMs including joint compound on gypsum board surfaces. No sampling is required in inaccessible portions of the crawl space, or at other inaccessible spaces.

The survey shall be conducted by an Asbestos Hazard Emergency Response Act (AHERA) Accredited Asbestos Inspector. Samples may be destructively taken, preferably from areas not highly visible. Apply coating or sealant to sampled areas to minimize fiber release.

Submit samples to an accredited laboratory for analytical testing for the presence of asbestos.

Perform a “point count” when the initial results indicate less than 2% asbestos.

Submit a report containing the results and recommendations. Include a floor plan showing the sampling locations and all chain of custody records.

Include a summary of the test results in the Project Specifications to assist the Contractor in bidding and in determining and adhering to appropriate standards for work procedures, testing, worker protection and environmental protection per OSHA and EPA requirements.

Abatement and disposal (or encapsulation if acceptable to the Government) shall be required.

13.2.2 Lead-Based Paint (LBP) Survey And Report

Analyze painted surfaces for lead content using a portable X-Ray Fluorescence (XRF) Analyzer at the locations that may be demolished, penetrated or otherwise disturbed during renovation. No sampling is required in spaces not readily accessible. NOTE: Sampling may be reduced to the extent that acceptable Government sampling and reports are available.

Samples may be destructively taken, preferably from areas not highly visible. Cosmetic repairs are required. Submit samples to an accredited laboratory for analytical testing for the presence of lead.

Paint on existing surfaces that contains more than 0.06 percent lead by weight of the non-volatile solids is considered LBP. [Part 35 of 24 CFR].

Submit a report containing the results and recommendations. Include a table or floor plan indicating the approximate sampling locations. Indicate precise locations of all positive samples. Include all chain of custody records.

Include a summary of the test results in the Project Specifications to assist the Contractor in bidding and in determining and adhering to appropriate standards for work procedures, testing, worker protection and environmental protection per OSHA and EPA requirements.

No abatement of LBP itself will be required. Encapsulation or removal of selected items with LBP may be required.

13.3 Pre-Design Planning

Prepare a Requirements and Management Plan (RAMP)/in accordance with instructions provided in individual TOs**.**

13.4 Design

13.4.1 Conceptual Design (10%)

Provide basic conceptual analysis including floor plans, elevations or renderings, room data sheets, site plans including above and below grade utilities, buildings vehicular and pedestrian circulation paths, parking, paved areas, conceptual landscape architectural concept, and existing site features to remain along with Anti-Terrorism Force Protection (ATFP) standoff distances. Include Fire Protection Code Compliance Summary Sheets. Provide outline specifications, in the form of a list of specification sections the Designer of Record (DOR) intends to use in the project. Provide the document as described in the UFC for design. Provide a listing of the UFGS used in the job by Section Number, Title, and Section Date. The Contractor shall provide a Rough Order of Magnitude (ROM) cost estimate at the conceptual stage. Estimate of time to complete construction of the project.

Provide conceptual alternatives for facility additions, alterations, new facilities and systems include options in finish materials, decor scheme, color, texture, and other project details. Specific guidance and required submittals will be provided in individual TOs.

13.4.2 Preliminary Design Phase (35%)

Typical Preliminary Design Submissions should include but not be limited to Basis for Design, Preliminary Drawings, Redlined Specifications, Preliminary Construction Cost Estimates and everything included from previous design submissions.

The Designer shall submit to the Government the following contract drawings, specifications, and other contract documents at this design stage addressing items defined in the TO, this document, and the more detailed Discipline-Specific UFCs, including documentation required by the UFC submittal requirements. All previous approved Design Charrette review comments shall be incorporated into this design. Include the following drawings/documents, with detail illustrated, defined by the Discipline-Specific UFCs and as applicable to the project:

Minutes of the Design Review Meeting and A-E's response to each of the Review Comments in electronic format.

13.4.2.1 Design Analysis Report

Submit a complete Design Analysis addressing items defined in this submittal and the Discipline-Specific UFCs including but not limited to: Preliminary load calculations and input/outputs per UFC requirements for mechanical, plumbing, fire protection, and electrical systems, and cybersecurity documentation per UFC 4-010-06 submittal requirements for all project included CE-related control systems including, but not limited to, Energy Management and Control Systems (EMCS), Fire Alarm Reporting System (FARS), Advanced Meter Reading System (AMRS), Emergency Generating Monitoring, etc.

13.4.2.2 Drawings

Submit a complete drawing set addressing items defined in this submittal and the Discipline-Specific UFCs. To include, but not limited to, the following:

a) Cover Sheet:
    a. Indicate Conceptual Design Phase.
b) Civil:
    a. Site Plan – Indicate project area footprint and limits of contract work, above and below grade utility lines, and contractor storage & staging area
    b. Site Features – Indicate buildings, parking, paved areas, preliminary landscape architectural concept, and existing site features to remain
c) Anti-Terrorism Force Protection (ATFP) – Standoff Distances
d) Landscape Architecture
    a. Preliminary landscape architectural concept
e) Architectural
    a. Floor Plans – Provide all floor plans, new and demolition, indicating room names and basic dimensions
    b. Building Elevations – Provide all building elevations indicating all exterior materials
    c. Building Sections – Indicate heights of critical building elements
    d. Room Data Sheets
f) Structural Drawings.
g) Mechanical
    a. Site Plan (if work outside of facility) - Show connection to base steam distribution, location of propane and oil tanks, layout of ground coupled heat pump well fields, etc.
    b. HVAC Floor Plan – Provide all floor plans, new and demolition, show equipment locations, one or two-line duct layout and preliminary piping runs.

    c. Mechanical Room Plan – Show major equipment and maintenance access space. Provide section view(s) to clarify layout and supports.
    d. Mechanical Schedules – show preliminary ventilation calculations, flammable refrigerant calculations, and major equipment schedules
    e. Preliminary facility related control system components, schematics, and points list.
    f. Preliminary Design Analysis including Preliminary load calculations and input/outputs per UFC requirements

h) Plumbing
    a. Floor Plan – Provide all floor plans, new and demolition, indicating room names, fixtures, floor drains, and equipment locations
    b. Isometrics
    c. Preliminary Design Analysis including Preliminary load calculations and input/outputs per UFC requirements

i) Fire Protection:
    a. Site Plan (if work outside of facility)
    b. Life Safety Floor Plan
    c. Code Compliance Summary Sheets
    d. Preliminary Design Analysis including Preliminary load calculations and input/outputs per UFC requirements

j) Electrical:
    a. Site Plan (if work outside of facility) - Provide all site plans, existing, new and demolition
    b. Electrical Systems Schematics
    c. Preliminary floor plans with dedicated space clearly identified for electrical and telecommunications rooms
    d. Preliminary facility related control system components, schematics, and points list.
    e. Preliminary Design Analysis including Preliminary load calculations for utility connections

13.4.2.3 Specifications

Submit specifications table of content, in the form of a list of specification sections the Designer of Record (DOR) intends to use in the project. Provide the document as described in the UFC for design. Provide a listing of the UFGS used in the job by Section Number, Title, and Section Date.

13.4.2.4 Schedule of Material Submittals

Submit a copy of the outline submittal register on the AF Form 66, Schedule of Material Submittals.

13.4.2.5 Conceptual Construction Cost Estimate

The Contractor shall provide a cost estimate using the standard Construction Specification Institute (CSI) format. The estimate shall show totals of each CSI division.

13.4.2.6 Conceptual Construction Schedule

Estimate of Construction Contract Time, including an initial critical path project schedule describing the construction phasing and primary construction work elements organized by the cost estimate CSI structure and narrative work schedule describing the construction activities for the purpose of pre-construction proposal evaluation.

13.4.3 Intermediate Design (65%)

Specific guidance and requirements will be provided in individual TOs. The intermediate design further refines the Preliminary Design and includes calculations, finishes, and costs. The drawings are typically about 65% complete at this stage. The submittal must include requirements of the previous submittal, including revisions due to review comments, plus additional detail to bring them to the required completion percentage.

The Designer shall submit to the Government the following copies of the contract drawings, specifications, and other contract documents at this design stage addressing items defined in the TO, this document, and the more detailed Discipline-Specific UFCs. All previous approved conceptual design review comments shall be incorporated into this design. Include not less than the following, as applicable to the TO:

Minutes of the Design Review Meeting and A-E's response to each of the Review Comments in electronic format.

13.4.3.1 Design Analysis Report

Submit a copy of the interim Design Analysis with edit notes (as necessary to document Preliminary Design Submittal reviews).

a) Geotechnical Report – Results of subsurface investigation – boring logs, test pit logs, etc. as an appendix.
b) Preliminary Calculations – Submit the following calculations, as a minimum, and as defined in detail by the Discipline-Specific UFCs.
c) Structural and Geotechnical – Submit Structural and Geotechnical calculations in sufficient detail to support the items outlined in the Design Analysis and indicated on the drawings.
d) Pavements – Submit Pavements calculations in sufficient detail to support the items outlined in the Design Analysis and indicated on the drawings.
e) Sustainable Design and Mechanical
    a. Energy Analysis - Submit a bound copy of the computerized energy analysis that includes input and output data in their entirety.
    b. Life Cycle Cost Analysis - Submit the computerized LCC analysis utilizing the latest edition of the NIST Building Life-Cycle Cost Program.
    c. Building Heating and Cooling Load - Submit a bound copy of the computerized load calculations with input and output data in their entirety.
    d. ASHRAE 90.1 Compliance Calculations - Submit calculations and compliance forms indicated in the Design Analysis.
    e. Cost Optimization/Reduction Analysis
f) Structural and Geotechnical – Submit Structural and Geotechnical calculations in sufficient detail to support the items outlined in the Design Analysis and indicated on the drawings.
g) Cybersecurity – Submit Cybersecurity documentation per UFC 4-010-06 submittal requirements for all project included CE-related control systems including, but not limited to, Energy Management and Control Systems (EMCS), Fire Alarm Reporting System (FARS), Advanced Meter Reading System (AMRS), Emergency Generating Monitoring, etc.
h) Electrical – Updated Design Analysis to substantiate design level shown. Submit calculations including
    a. Lighting: Interior and Exterior Foot-candles
    b. Load Analysis
    c. Service size
    d. Feeder size
    e. Generator Size

    f. Larger special circuit sizes
    g. Lightning Risk Assessment
- i) Fire Protection – Submit all calculations supporting all fire suppression and fire alarm/detection systems for the project. Calculations for systems, features, or elements other than fire suppression or detection will be required as applicable. Fire suppression system calculations must be prepared using commercially available computer software.
- j) Preliminary Environmental Report for "For Construction" Design – Submit reports as required in UFC 3-800-10N.

13.4.3.2 Drawings

Include the following drawings as defined in the Discipline-Specific UFCs and as applicable to the project, to include in addition to the conceptual requirements; but not limited to, the following:

- a) Cover and supplement sheet
    - a. Indicate Preliminary Design Phase
    - b. Show Base Map with Project Location, Contractors' Haul Route, Base Entrances and Key Buildings
    - c. Vicinity and Project Location Maps
    - d. Index of Drawings
- b) Civil
    - a. Legend and Notes
    - b. Utility Plan - Include cathodic protection for utilities
    - c. Layout Plan for Roads and Parking – Indicate vehicular and pedestrian circulation paths
    - d. Datum security tied between National Geodetic Vertical Datum (NGVD) and local datums
    - e. Pavement sections – Including joint layout plan and details.
    - f. Show building footprint
    - g. Prominent site features – Including fences (if any), grading, and drainage. Use 2D polyline with elevations for existing and new contour lines and finish grades.
- c) Landscape Architecture
    - a. Planting Plan – show locations of all facilities (buildings, parking areas, roads, existing vegetation noted for preservation, etc.) and new plantings (trees, shrubs, ground cover, etc.).
    - b. Plant Schedule and Details – provide a schedule for plant material showing as a minimum: common name, botanical name, quantity of plants, root condition (balled and burlapped, containerized, boxed, etc.), and a keyed reference to a planting detail. Provide separate details for plant types (trees, shrubs, ground covers, etc.).
    - c. Miscellaneous Plans and Details – provide plan drawings and details for specialized construction including such items as plazas, courtyards, child play equipment, monuments, memorials, site furniture, fences, walls, signage, etc.
    - d. Irrigation Plan – show all water lines, sprinkler heads, valves, backflow preventers, water source connections, wells, automatic controllers, schedules, etc. when a site irrigation system is required.
    - e. Irrigation Details – when a site irrigation plan is required, provide details of sprinkler heads, backflow preventers, valves, accessories, etc.
- d) Asbestos and hazardous materials abatement (if required)
    - a. Show locations of all asbestos-containing-materials (ACM) scheduled for abatement and all ACM to remain. (Show on separate drawing, NOT with general demolition).
    - b. Show locations of Lead-Based paint (LBP) on the Drawings to assist the Contractor in determining and adhering to appropriate standards for work procedures, testing, worker protection and environmental protection per OSHA and EPA requirements. LBP

abatement is NOT anticipated. (Show on separate drawing, NOT with general demolition).

e) Architectural
   a. Floor Plans – Provide all floor plans, new and demolition, indicating room names and dimensions, Ceiling Plan, Furniture Plan
   b. Building Elevations – Provide all building elevations indicating all exterior materials
   c. Roof Plan – Provide a plan of all roof areas, indicating direction of slope and method of drainage
   d. Building Section – Indicate heights
   e. Typical Wall Sections – Provide sufficient wall section(s) to indicate all materials and different conditions
   f. Views (doors, windows, cabinets, interiors)
   g. Schedules – Indicate all proposed schedules (doors, hardware, windows, finishes)
   h. Details
f) Structural
   a. Foundation Plans - Include for all structures, showing dimensions, arrangements, elevations, locations referred to a column line grid system, type of foundation and foundation obstructions. Include the layout of all slabs, footings, piers, grade beams, piles, etc., showing all foundation features of the design.
   b. Framing Plans – Include a framing plan for each structural level of the facility, showing dimensions, elevations, and column locations and numbering referenced to a column line grid system, and overall sizes of major members and components. Show the layout of beams, joists, stringers, etc.
   c. Structural Details – Show typical details of construction, indicating the connection and relationship between major components of the structural system.
   d. Structural Elevations – Show general sizes, location and arrangement of all significant features of the vertical framing system, including columns, walls, beams, etc.
g) Mechanical
   a. Site Plan (if work outside of facility) - Show connection to base steam distribution, location of propane and oil tanks, layout of ground coupled heat pump well fields, etc.
   b. HVAC Floor Plan – Show equipment locations, one or two-line duct layout and preliminary piping runs.
   c. Mechanical Room Plan – Show major equipment and maintenance access space. Provide section view(s) to clarify layout and supports.
   d. Sections
   e. Details (Show layouts for plumbing, HVAC, EMCS, and fire suppression systems. Indicate all primary and terminal equipment, piping including sizes, valves, ductwork including sizes, water and air flows.)
h) Plumbing
   a. Site Plan (if work outside of facility)
   b. Floor Plan – Indicate fixtures, floor drains, and equipment locations
   c. Details
   d. Schedules
i) Fire Protection
   a. Code Compliance Summary Sheets (Updated from Concept Design Submittal)
   b. Life Safety plan (Updated from Concept Design Submittal)
   c. Floor Plans – Provide all floor plans, new and demolition, indicating room names. Include Fire Suppression plans, and Fire Alarm and Mass Notification System Plans
   d. Detail Sheets – Include Fire Alarm Sequence and Riser Diagrams
j) Electrical
   a. Legend and Abbreviations

b. Floor Plans – Provide all floor plans, existing, new and demolition, indicating room names, indicate layouts for power including sizes of components, lighting, cathodic protection, communications (including LAN), fire detection, and other special systems.
c. Lighting Plans and Details
d. Power Plans and Details
e. Power - Single Line Diagram
f. Additional Plans/Risers - To include cathodic protection, communications (including LAN), fire detection, telephone, and other special systems

k) Telecommunications (registered communications distribution designer (RCDD) approved)
  a. Telecommunication Drawings:
    i. T1 - Layout of complete building per floor
    ii. T2 - Serving Zones/Building Area Drawings
    iii. T4 - Typical Detail Drawings
  b. Telecommunications Space Drawings
    i. T3 - telecommunications rooms plan views, pathway layout (cable tray, racks, ladder-racks, etc.), mechanical/electrical layout, and cabinets, racks, backboard, and wall elevations.
  c. Telecommunications Qualifications:
    i. Energy Analysis - Provide a bound copy of the computerized energy analysis that includes input and output data in their entirety.
    ii. Life Cycle Cost Analysis - Submit the computerized LCC analysis utilizing the latest edition of the NIST Building Life-Cycle Cost Program.
    iii. Building Heating and Cooling Load - Provide a bound copy of the computerized load calculations with input and output data in their entirety.
    iv. ASHRAE 90.1 Compliance Calculations - Submit calculations and compliance forms indicated in the Design Analysis.
  d. Test Plan

l) Security Systems
  a. The designer must show sensor detection patterns, entry control terminal devices, portal control, facility interface, personnel identity verification, surveillance and detection equipment locations, and quantities and installation details on drawings. Requirements for additional site-specific conditions may include furniture/equipment layout within protected areas, and hazard location areas, type of hazard, class, and group. See UFGS-28 20 01.00 10 for details.

13.4.3.3 Specifications

Submit outline specifications, in the form of a list of specification sections the Designer of Record (DOR) intends to use in the project. Provide the document as described in the UFC for design. Provide a listing of the UFGS used in the job by Section Number, Title, and Section Date.

13.4.3.4 Schedule of Material Submittals

Submit a draft submittal register on the AF Form 66, Schedule of Material Submittals with edit notes (as necessary to document 35% Design Submittal reviews).

13.4.3.5 Preliminary Construction Cost Estimate

Submit a detailed Construction Cost Estimate Breakdown in the most current standardized Construction Specification Institute (CSI) MasterFormat structure utilizing the AF Form 3052, Construction Cost Estimate for the construction costs associated with the design project.

13.4.3.6 Preliminary Construction Schedule

Estimate of Construction Contract Time, including an initial critical path project schedule describing the construction phasing and primary construction work elements organized by the cost estimate CSI structure and narrative WBS describing the construction activities for the purpose of pre-construction proposal evaluation.

13.4.4 Advanced Final Design (95%) and Final Design Submittal (100%)

The Advanced Final Design Development Submittal is intended to convey the complete extent of the work in a final manner. The Designer shall submit to the Government the following copies of the contract drawings, specifications, and other contract documents at this design stage addressing items defined in the TO, this document, and the more detailed Discipline-Specific UFCs. Based upon all previous data and approved design review comments, this submittal shall be the Contractor's best attempt at 100% design. Ideally, no corrections or additional design work will be required. Include not less than the following, as applicable to the TO:

Minutes of the Design Review Meeting and Contractor's response to each of the Review Comments in electronic format.

One compact disk (CD) with PDF copies of all documents (Specifications/AF Form 3052/AF Form 66) and PDF and AutoCAD copies of all drawings.

13.4.4.1 Design Analysis Report

Submit revised Design Analysis including updated information and incorporating responses to previous government review comments.

a) The Geotechnical Report, if modified during the previous review, shall be re-submitted as an appendix to the Design Analysis; otherwise, do not submit.
b) Cybersecurity – Submit Cybersecurity documentation shall be resubmitted in full and conform to UFC 4-010-06 submittal requirements for all project included CE-related control systems including, but not limited to, Energy Management and Control Systems (EMCS), Fire Alarm Reporting System (FARS), Advanced Meter Reading System (AMRS), Emergency Generating Monitoring, etc.
c) Pre-Final Calculations – Revise calculations as required to reflect resolution of all previous government review comments. Provide design analysis that is 100% complete. Provide calculations for each discipline in addition to the following detailed requirements.
    a. Mechanical – Submit calculations to support the plumbing and mechanical systems and the major equipment comprising those systems. Submittals shall include, but not be limited to cooling loads, heating loads, air balance, and outside air calculations. Update the energy analysis, provided at the Preliminary phase, with the equipment efficiencies scheduled on the drawings.
    b. Electrical – Provide updated Calculations from previous submittal to substantiate design level shown, including the following, as applicable.
        i. Short Circuit
        ii. Voltage Drop
        iii. Lighting
        iv. Load Analysis.
        v. Generator sizing calculations
        vi. Motor Starting/Flicker Analysis
        vii. Sag, Tension, and Guying Analysis
        viii. Manhole Design Calculations

ix. Cable Pulling Tension Calculations
x. Cathodic Protection Calculations
xi. CATV Network Loss Calculations

13.4.4.2 Drawings

Drawings must be 100% complete, minus final signatures, and modified to reflect the responses to previous review comments. The drawings must be complete to the extent that they may be released for bid or constructed as submitted to include in addition to the preliminary requirements; but not limited to, the following:

a) Cover and supplement sheet
    a. Indicate 100% Design Phase
    b. Complete including Abbreviations, Architectural Legends, Architectural Symbols, Architectural Material Designations
    c. Schedule of any Bid Additives (or deductives) necessary to bring the project in within budget
b) Civil
    a. Complete with all legends, notes, schedules
    b. Profile sections – Provide utility and street, indicate points of intersection (PI) for all curves.
    c. Grade elevations
    d. Sufficient information to meet State of Arizona or Pima County DEQ requirements, when required
c) Landscape Architecture
    a. Complete with all legends, notes, schedules
d) Abatement
    a. Complete with all notes
e) Architectural
    a. Complete with all legends, notes and schedules
f) Structural Drawings
    a. Complete with all notes and schedules
g) Mechanical
    a. Complete with all legends, notes and schedules including equipment and fixture schedules
    b. Plumbing riser diagrams
    c. HVAC control drawings, to include cybersecurity documentation requirements
h) Plumbing
    a. Complete with all legends, notes and schedules including equipment and fixture schedules, and plumbing riser diagrams
i) Fire Protection
    a. Complete with all legends, notes and schedules
    b. Provide, as a minimum, all noted drawings in UFC 3-600-10N in addition to drawing requirements specified for Design Development.
    c. Fire Alarm Riser Diagram – Include only when separate Fire Protection Drawings are not required to be included in the design.
        i. Other Riser Diagrams for Television, Security, and similar systems
        ii. Panel Schedules
        iii. Switchboards and Motor Control Center Schedules
        iv. Lighting Fixture Details
j) Electrical

    a. Complete with all legends, notes and schedules
    b. Lighting Fixture Schedules
    c. Panel Board Schedules
    d. Telephone Riser Diagram
    e. Intercommunication Riser Diagram
    f. Lightning Risk Assessment

k) Telecommunications (registered communications distribution designer (RCDD) approved)
    a. Complete with all legends, notes and schedules

l) Security Systems
    a. Complete with all legends, notes and schedules

13.4.4.3 Specifications

Pre-Final submittal should be complete at this stage and require only minor corrections if any, including complete description of any Bid Options (additive or deductive) necessary to bring the project within budget.

For Fire Protection systems only, provide manufacturer's data sheets instead of prescriptive specifications, for the suppression systems, the detection and alarm systems, firestopping, and spray-applied fireproofing.

13.4.4.4 Schedule of Material Submittals

Submit the final submittal register on the AF Form 66, Schedule of Material Submittals.

13.4.4.5 Pre-Final Construction Cost Estimate

Revised detailed Construction Cost Estimate Breakdown that reflects resolution of all previous government review comments utilizing the AF Form 3052, Construction Cost Estimate.

13.4.4.6 Pre-Final Construction Schedule

Revised schedule that reflects resolution of all previous government review comments.

13.4.4.7 Color Board

Submit separate interior and exterior color documentation binders indicating all proposed material and color selections.

13.4.5 Constructability Review Certification

The certification shall be signed by the Contractor's PM and by all technical consultants used for review of this project. Provide the signed certification document with the Final Design Submittal. Specific guidance and required submittals will be provided in individual TOs.

13.4.6 Color Renderings

When specified, the AE shall prepare full color rendering illustrating an exterior view of the project building and site development. The rendering shall be a bird's eye view and illustrate in an accurate manner, significant architectural features of the proposed project. The renderings shall clearly indicate the project title, location and fiscal year in the lower right-hand corner. The following formats are to be provided as part of this work:

13.4.6.1 Framed Original & Electronic Renderings

If required, will be specified in the subsequent Task Orders and proposal requests.

13.4.7 Submittal Review

Reviewing officials and length of time required for review will be determined based on the level of effort and specified with each Task Order.

13.5 Project Support – Pre-Proposal/Bid Through Construction

13.5.1 Preparation of Bid Schedule

Coordinate the bid schedule with the TO PM and include a complete list of bid items, Base Bid and Options, with instructions for bidding and award.

13.5.2 Pre-Proposal/Bid Opening

Provide clarifications to questions regarding the construction documents within **5 workdays** to the TO PMunless agreed to otherwise or specified in individual TOs.

13.5.3 Proposal/Bid-Opening

Provide recommendations, and correct/adjust the project design, to include drawings, specifications, design analysis, and cost estimates as specified in individual TOs.

13.5.4 Construction Support

Respond to job site concerns and prepare addenda, change orders, and related cost estimates during construction.

13.6 Support for Design-Build (D-B)

Scope project to support RFP development, to include, preparation of Statement of Work (SOW) performance specifications, independent cost estimate, and drawings as specified in individual TOs. The balance of design and construction technology is determined by the D-B team. Identify key elements to be submitted by offerors such as a management plan and technical design solutions. The Conceptual Design is intended to convey the extent of the work in a preliminary manner.

13.6.1 Charrette for Conceptual Design

The Contractor shall attend a Charrette. The Charrette will be a cooperative effort by the design team, user and client representatives, facility engineering command personnel, and other interested parties for the purpose of validating and clarifying the project requirements. The Charrette will include on-site development of a conceptual design in response to functional, aesthetic, environmental, base planning, site, budgetary, and other requirements. At the end of the Charrette, the Contractor will provide an out-brief consisting of meeting minutes, decision documentation, and information that led up to elements that are, or will be, included in the conceptual design.

The Charrette out-brief will detail a narrative presentation of facts sufficiently complete to demonstrate that the project concept is fully understood and that subsequent design details and their ultimate presentation in the final submittal will be based on sound architectural and engineering decisions. The narrative will address Federal High Performance and Sustainable Buildings Guiding Principles status, and

LEED™ status. The narrative shall be detailed enough to use as the Scope of Work for a D-B Request for Proposal (RFP). Provide a discussion and description of the design in each of the disciplines appropriate to the project.

13.6.2 Conceptual Design

The conceptual design shall include the following elements defined by the Discipline-Specific UFCs and as applicable to the project.

13.6.2.1 Conceptual Design Analysis

Submit a complete design analysis addressing items defined in this submittal and the Discipline-Specific UFCs.

13.6.2.2 Architectural Drawings

a) Floor Plans: Provide all floor plans, new and demolition, indicating room names and basic dimensions.
b) Building Elevation: Provide all building elevations indicating all exterior materials.
c) Building Section: Indicate heights of critical building elements.
d) Room Data Sheets

13.6.2.3 Civil Drawings

Conceptual Site Plan

Indicate above and below grade utility lines, vehicular and pedestrian circulation paths, buildings, parking, paved areas, preliminary landscape architectural concept, and existing site features to remain.

Anti-Terrorism Force Protection (ATFP) Standoff Distances

13.6.2.4 Electrical Drawings

Need not provide extensive details but must be complete enough to thoroughly express the Designer's intentions and include the following:

a) Existing Site and Demolition Plan
b) Site Plan
c) Single Line Diagram
d) Preliminary floor plans with dedicated space clearly identified for electrical and telecommunications rooms
e) Preliminary Design Analysis including Preliminary load calculations for utility connections
f) Preliminary Design Analysis including Preliminary load calculations for utility connections

13.6.2.5 Fire Protection Drawings

a) Code Compliance Summary Sheets
b) Life Safety Floor Plan

13.6.2.6 Plumbing Drawings

Plumbing Floor Plan. Show fixtures, floor drains, and equipment locations.

13.6.2.7 Mechanical Drawings

a) Site Plan. Show connection to base steam distribution, location of propane and oil tanks, layout of ground coupled heat pump well fields, etc.
b) HVAC Floor Plan. Show equipment locations, one or two-line duct layout and preliminary piping runs.
c) Mechanical Room Plan. Show major equipment and maintenance access space. Provide section view(s) to clarify layout and supports.

13.6.2.8 Communications and Security Systems

Discuss how designs, and design analysis meet the requirements of UFC 3-580-01, Telecommunications Building Cabling Systems Planning and Design, ETL 02-12, Communications and Information System Criteria for Air Force Facilities, NSTISSAM TEMPEST/2-95, NSTISSI 7003, Protected Distribution Systems, AFI 31-101, Integrated Defense, and ICD 705, Sensitive Compartmented Information Facilities (As necessary)

Discuss how designs, and design analysis meet the requirements of UFC 4-020-01, DoD Security Engineering Facilities Planning Manual, UFC 4-020-03FA, Security Engineering: Final Design, AFI 31-101, Integrated Defense, and ICD 705, Sensitive Compartmented Information Facilities (As necessary).

a) Design criteria as the basis for defining a protective system that mitigates vulnerabilities to assets.
b) The criteria describe the assets associated with a facility, the threat to those assets, the level to which those assets are to be protected against the threat, and any constraints to the protective system design that may be imposed by the Planning Team.
c) For existing facilities, vulnerabilities as additional factors in establishing the design criteria. Those vulnerabilities will be based on evaluating how existing conditions affect the protection of the identified assets against the identified threats to the applicable levels of protection.

13.6.2.9 Cybersecurity for Facility Related Control Systems

Discuss how designs, and design analysis meet the requirements of UFC 4-010-06, Cybersecurity of Facility Related Control Systems

a) Cybersecurity – Submit Cybersecurity documentation shall be resubmitted in full and conform to UFC 4-010-06 submittal requirements for all project included CE-related control systems including, but not limited to, Energy Management and Control Systems (EMCS), Fire Alarm Reporting System (FARS), Advanced Meter Reading System (AMRS), Emergency Generating Monitoring, etc.

13.6.2.10 Geotechnical Report

Include the Geotechnical Report as an appendix.

13.6.2.11 Conceptual Outline Specifications

Provide outline specifications, in the form of a list of specification sections intended for the project. Provide the document as described in the UFC for design. Provide a listing of the UFGS used in the job by Section Number, Title, and Section Date.

13.6.2.12 Conceptual Design Cost Estimate

The Contractor shall provide a cost estimate using the standard Construction Specification Institute (CSI) format. The estimate shall show totals of each CSI division.

13.6.2.13 Conceptual Design Schedule

Estimate of Conceptual Design Time, including an initial critical path project schedule describing the design phasing and primary design work elements organized by the cost estimate CSI structure and narrative work schedule describing the construction activities for the purpose of pre-construction proposal evaluation.

14.0 TITLE II, CONSTRUCTION SERVICES

Title II, Construction Services are related to a specific construction project and consist of inspection and documentation of construction. Title II services may include all aspects of construction quality assurance and oversight of facility and infrastructure construction/renovation projects. The Contractor shall provide sufficient on-site expertise during the contract to assure itself and the Base Civil Engineer that adequate periodic observation of the construction work for contract compliance is being performed. The Contractor shall provide the level of effort as indicated in TO.

Furnish personnel who are professionally qualified by nature of education and experience to perform all duties required for construction services. This qualification requirement extends to all consultants that may be retained by the Contractor for purposes of performing Title II Construction Services. Resumes shall be submitted for verification of expertise in required field. These individuals may be the same person if credentials validate their knowledge in the multiple fields.

Evaluate all test reports, coordinate evaluations with the project manager and/or construction inspector and report to the Contracting Officer in writing, as to conformity or nonconformity of the materials tested with the specifications.

Check shop drawings, samples and materials submittals for compliance with all contract documents.

Provide daily inspection services and document them on AF Form 1477 during the construction performance period and a survey party to check locations and elevations with a written report on the survey results. The Contracting Officer will have overall jurisdiction covering the surveillance, inspection, and administration during the construction of the project. These services shall be furnished in accordance with the requirements set forth in the TO placed with the Contractor for Title II services.

Notify the Contracting Officer, in writing, of any modifications deemed necessary or required as a result of shop drawings or materials submittals reviews for improvement of the completed project. Adequate justifications and/or engineering analysis of modification plus cost estimates, shall accompany any recommendations for changes to the drawings and/or specifications.

Approved changes to the contract that result from the Contractor's review of shop drawings and other submittals shall be annotated on one copy of the drawings and specifications by the Contractor. Upon completion of the contract, copies of all submittals as well as the above marked copy of the drawings and specifications shall be forwarded to the Contracting Officer.

14.1 Limitations on Authority of Contractor.

Unless specific exceptions are established by a written instruction issued by the C.O., the Contractor:

- Shall not authorize any deviation from the construction contract documents or approve any substitute materials or equipment.
- Shall not exceed limitations on the government's authority as set forth in construction contract documents.
- Shall not undertake any of the responsibilities of construction Contractors, subContractors, construction Contractor's superintendent or Contractor quality control representatives.
- Shall not expedite or accelerate the work of construction Contractors and subContractors.
- Shall not advise on or issue directions relative to any aspect of the means, methods, techniques, sequences, or procedures of construction unless such is specifically called for in construction contract documents.
- Shall not authorize or advise users to occupy projects in whole or in part.

15.0 OTHER SERVICES

Provide A-E services of interest to the Government not necessarily connected with a specific construction projects or programs. Other services may include:

15.1 Meetings and Teleconferences

15.2 Meeting/Teleconference Support

Attend meetings and teleconferences as required by the Contracting Officer's Administrator (CA) with the TO project manager, customer, and government representative(s). These will be held at Davis-Monthan AFB, AZ. The purpose of the meetings includes, but is not limited to, contract discussions, progress reviews, design reviews, planning/programming charrette activities, design charrette activities, project status, and the general exchange of information concerning current and future activities.

Coordination conferences or meetings may be held as requested by either the Contractor or the Air Force.

The Contractor shall prepare written minutes or memoranda of all design review meetings and of pertinent instructions, and information received verbally or otherwise. Copies of minutes or memoranda in electronic format shall be transmitted to the Base Contracting Officer within fourteen (14) days of the design review meeting.

Provide response in electronic format to each of the Government's design review comments no later than the date of the next design submittal.

The entire log of minutes, memoranda, or review comment response shall be included in the Design Analysis submission at each design phase.

15.3 Predesign/Charrette In-Brief Meeting

The Contractor shall attend a Charrette. The Charrette will be a cooperative effort by the design team, user and client representatives, and other interested parties for the purpose of validating and clarifying the project requirements. The Charrette will include on-site development of a conceptual design in response to functional, aesthetic, environmental, base planning, site, budgetary, and other requirements. At the end of the Charrette, the Contractor will provide an out-brief consisting of meeting minutes, decision documentation, and information that led up to elements that are, or will be, included in the conceptual design.

The Charrette out-brief will detail a narrative presentation of facts sufficiently complete to demonstrate that the project concept is fully understood and that subsequent design details and their ultimate presentation in the final submittal will be based on sound architectural and engineering decisions. The narrative shall be detailed enough to use as the Scope of Work for a D-B Request for Proposal (RFP). Provide a discussion and description of the design in each of the disciplines appropriate to the project.

15.4 Digital Imaging

Provide digital imaging with at least 1536 x 1024 pixels and 32-bit color as specified in the TOs.

15.5 Project Administrative Record

The Project Administrative Records shall be saved as a “searchable” .pdf format on CD-ROM or DVD and include all documentation, correspondence, personal contacts, studies, surveys, and cited references used in the preparation of the final deliverable, and information/correspondence provided by Government personnel and participants/representatives. Specific deliverables will be identified in the TO.

15.6 Conference and Regulatory Agency Interaction Support

15.6.1 Conference Support

Provide support in arranging, scheduling, developing agenda, programs, preparing minutes, perform training sessions, and other related support as specified in individual TOs.

15.6.2 Regulatory Agency Interaction Support

Provide support in arranging, scheduling, developing agenda, news releases, notices, programs, minutes, proceedings, and other related support required when interacting with regulatory or other professional agencies as specified in individual TOs.

15.7 Graphics Support

Provide graphic design, production and printing oversight of various items in support of various project activities as specified in the TOs.

15.8 Publication Development and Production

Provide research, production (text, graphics and/or photographic designs), and website and/or hardcopy development or revision for various Government regulations, standards, publications, handbooks, pamphlets, and/or guides as specified in individual TOs as specified in individual TOs.

15.9 Comprehensive Interior Design (CID)

Provide CID support, including designing, selecting, and developing the interior building finishes, special effects, and furnishings for an integrated visual design theme as specified in individual TOs. Furnishings include loose furniture, systems furniture, graphics, wall hangings, waste receptacles, chalk and tack boards, refrigerators, ranges, microwaves, entry directories, and other similar items. Requested estimates and products may be required to be sourced/specified from mandatory use contracts.

15.10 Technical Investigation

Perform field and topographic surveys, utility location and capacity analysis, toxic and hazardous material surveys, geotechnical investigations, and interior and exterior concept studies. Perform field investigations and research necessary to ascertain all existing conditions affecting the design and project construction as specified in individual TOs.

15.11 Cost Estimates

Develop detailed cost estimate following the Contruction Specifications Institute (CSI) format utilizing RSMeans; Micro Computer Aided Cost Estimating System (MCACES); or equivalent parametric cost estimating system as specified in individual TOs.

15.12 Operations and Maintenance (O&M) Plan

Develop an O&M Plan as specified in individual TOs.